Learning Physics from Noisy Data

Nonlinear pendulum dynamics and formal verification in Lean 4

Research question. Can an interpretable model learned from noisy angle measurements predict unseen motion and satisfy the underlying energy law?

244 frozen equations	732 held-out rollouts	13 audited Lean theorems
Sparse fits across two training sets and two estimators	Tests at 45°, 75° and 90°	Exact rate identities and interval conservation

Approach

I used a frictionless pendulum as a benchmark with a known law. After comparing linear, sinusoidal and cubic baselines, I used sparse identification of nonlinear dynamics (SINDy) to fit equations from angles with Gaussian noise up to 5° SD. I froze selection before testing unseen initial conditions and used Lean 4 to connect the learned acceleration to physical energy.

What the results show

Learning from noise. Savitzky-Golay preprocessing recovered the sine-only structure in 31 of 120 noisy fits, compared with 0 of 120 for central differences. At 5° noise with the broader training set, mean held-out angle RMSE fell from 1.255 to 0.683 rad across 20 matched repetitions. The sine function was supplied in the candidate library.

Diagnosing failure. Of 732 rollouts, 98 hit the speed guard under both solver settings. In one case a model with validation error of 0.0407 rad predicted outward acceleration at a 90° release, where gravity should restore the pendulum. The discrepancy exposed a failure of extrapolation from small-angle validation.

Connecting learning and proof. Lean proves the general rate $dE/dt = \omega[f(\theta, \omega) + k \sin \theta]$ and ideal interval conservation. The linear baseline at 90° conserved its own model energy to numerical precision while its true-reference energy varied by about 20%. This is why trajectory accuracy, energy consistency and formal claims were assessed separately.

Research direction

This benchmark uses synthetic data. I would next broaden validation amplitudes, compare integral identification and test energy constraints on independently measured trajectories.

Reproducible record. The companion project contains the full research report, Lean proofs, frozen model coefficients and experiment records.